APPENDIX II. DEBT SUSTAINABILITY ANALYSIS (UPDATE)

This appendix updates the joint debt sustainability analysis (DSA) from April 2008. Overall, the debt sustainability outlook remain largely unchanged. Burkina Faso is classified as high risk of debt distress because the threshold for the NPV of debt-to-exports ratio is breached in the medium term.

I. EXTERNAL DEBT SUSTAINABILITY ANALYSIS

A. What Has Changed Since the Last DSA?

1. **The medium-term economic outlook remains broadly unchanged**. Projections for real GDP growth remain unchanged over the medium term. However, the surge in inflation in 2008 is expected to have a lasting impact on the price level. As a result, nominal GDP projections are somewhat higher. Export diversification and structural reforms implementation remain key to sustain growth over the medium term (Figure 1).

2. **Fiscal developments are broadly in line with the previous DSA**. The authorities remain committed to a prudent fiscal policy and a tax policy reform is underway. Projections for new borrowing in US dollars are lower because of the appreciation of the dollar.

Box 1. Burkina Faso—Macroeconomic Assumptions

Real GDP growth is projected to average 6 percent in the medium and long run, broadly in line with the historical average of the past ten years. Growth in 2008 and 2009 is projected to be below trend as a result of exogenous shocks (weather conditions, external environment). Over the medium and long term, growth is expected to reach 6 percent, as a result of improvements in productivity and export diversification.

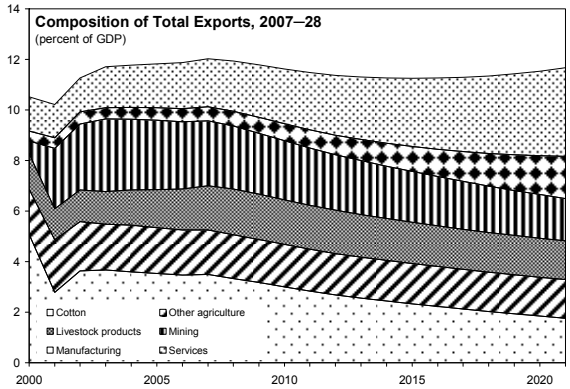
Inflation is expected to decline gradually as food and oil prices pressures ease, and is expected to return to its historical average of about 2 percent over the medium term.

The **overall fiscal deficit**, with and without grants, is projected to decrease over time as revenue increases towards the WAEMU tax revenue target of 17 percent of GDP. The revenue increase is supported by ongoing efforts to strengthen tax and customs administration, which is a centerpiece of the current Fund-supported program, and comprehensive tax policy reforms. Expenditure is projected to remain high to meet the authorities' PRSP objectives. External financing requirements are to be met equally by grants and loans, which is a conservative assumption based on recent experience.

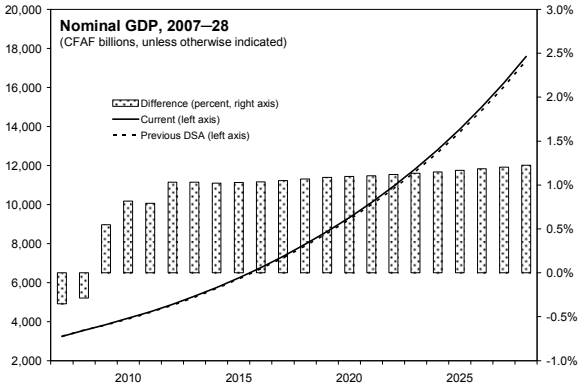
The **current account deficit** is expected to narrow in the medium and long run as the tighter fiscal stance dampens import demand. The exports-to-GDP ratio will stabilize at about 10 percent of GDP over the medium term. Export diversification (gold, manufacturing goods, services) will be key to achieving growth objectives.

Figure 1. Burkina Faso: Medium-Term Framework

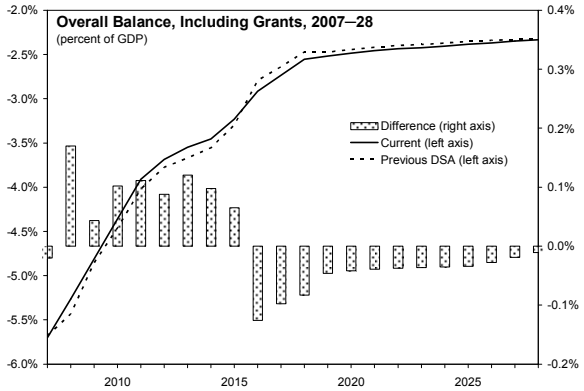
Exports are expected to be more diversified over the medium term.

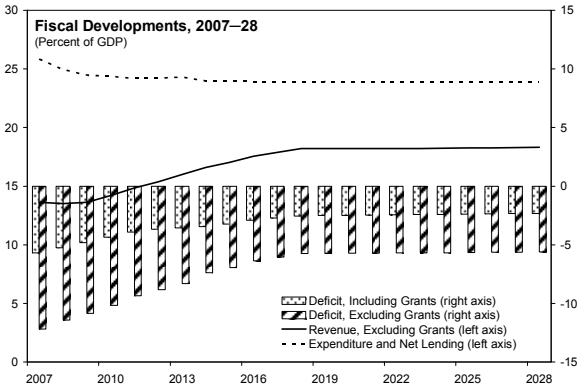


Nominal GDP is projected to be somewhat higher, largely because of price developments.

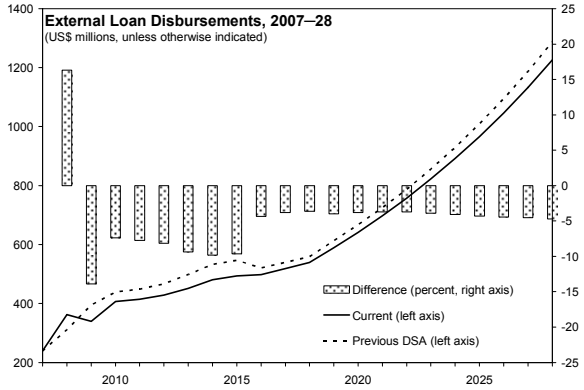


Revenue is expected to increase, as a result of the implementation of fiscal reforms, which will contribute to reducing the overall fiscal deficit over the medium term.

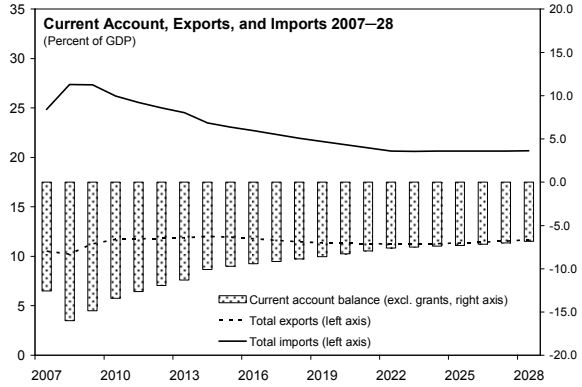




Borrowing requirements are lower as a result of a slightly stronger fiscal adjustment over the medium term.



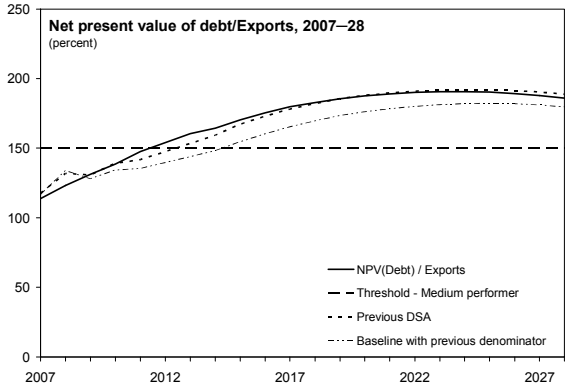
Current account balance is expected to narrow as tighter fiscal stance dampens imports demand.



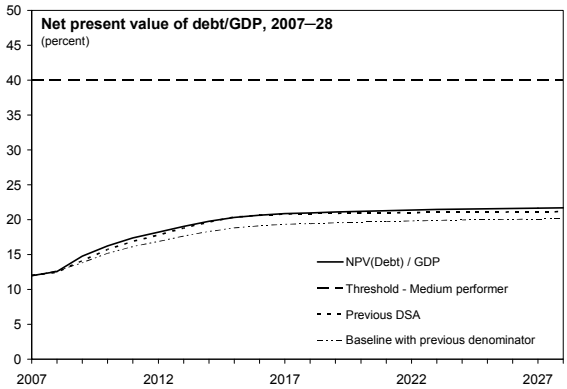
Sources: Burkinabè authorities, World Economic Outlook, and IMF staff estimates and projections.

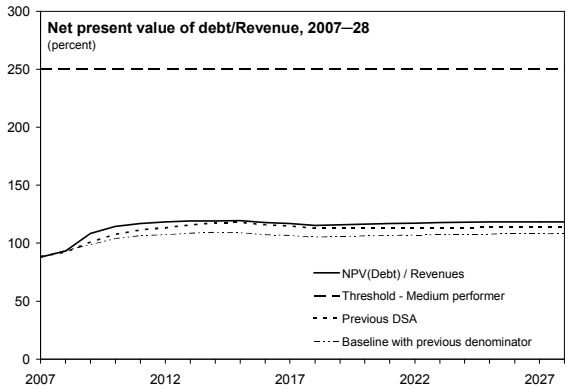
Figure 2. Burkina Faso: Current vs. Previous DSA

The main risk remains the NPV of debt to exports ratio.

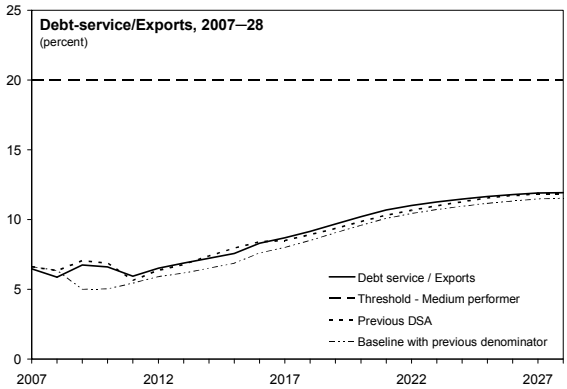


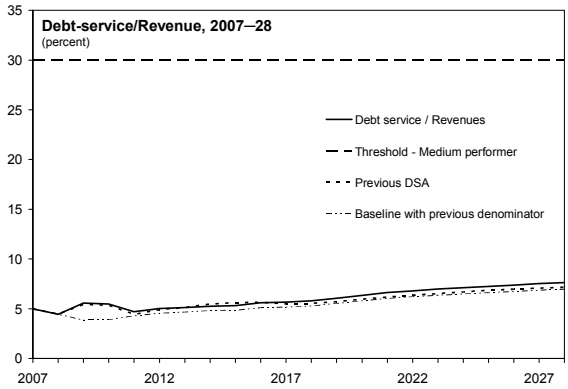
NPV of debt to GDP and revenue ratios are below the indicative thresholds.





Debt service ratios are also well below thresholds.





Sources: Burkinabè authorities, World Economic Outlook, and IMF staff estimates and projections.

B. Assessing Vulnerabilities

3. **A worse-than-expected export and growth performance is a clear risk**. Burkina Faso remains heavily dependent on cotton production and exports, although the projected increase in gold production should help diversify exports. A clear risk could occur in case of either a sharp fall in cotton and gold prices or disruptions in cotton production.

4. **A continuous decline in the fiscal deficit throughout the projection period is important for the resilience of debt indicators to an adverse export or growth shock**. This implicitly assumes that (i) structural reforms in the area of tax administration and fiscal policy would be implemented and (ii) that fiscal tightening would be adopted as a response to permanent losses of revenue.

5. **There are a number of other positive factors as well as some downside risks:**

Positive factors

- Burkina Faso's government has consistently demonstrated its commitment to stay current on its external payment obligations. This includes the period before the country benefited from debt relief under the Heavily Indebted Poor Country Initiative (HIPC), when the NPV of debt-to-exports ratio was well above 200 percent.
- Additional grant financing could improve the debt sustainability outlook significantly.
- The authorities have adopted a national debt policy and received technical assistance in this area from the World Bank.
- Improvements in CPIA rating, which depends on the implementation of several key structural reforms (fiscal, financial sector, business climate), could bring Burkina Faso back to a classification as a strong performer, which would induce higher thresholds.

Downside risks

- Delayed fiscal adjustment could pose a significant downside risk. This is because without deficit reduction Burkina Faso's debt burden could eventually become unsustainable, even when measured relative to GDP or revenue. Preventing this will require a stronger revenue effort, which in turn depends on further efforts to strengthen tax and customs administration, and to implement substantial tax policy reform.
- Adverse export shocks would mostly affect the export-based debt indicators, whereas the other indicators are relatively robust with respect to this type of shock.

C. Risk of Debt Distress

6. **The risk of debt distress is high because the threshold for the NPV of debt to exports ratio is breached**. The peak ratio of about 190 percent in the baseline projection of the NPV of debt-to-exports ratio is well above the 150 percent threshold. Under DSF guidelines, such a breach warrants a classification of high risk of debt distress, although the GDP- and revenue-based DSF indicators remain well below their indicative thresholds, including some of the alternative scenarios and stress tests.

II. Fiscal debt sustainability analysis

7. **The results for the fiscal DSA are broadly similar to those of the external DSA**. Failure to reduce the current fiscal deficits would lead to ever-increasing debt indicators, which is clearly unsustainable. The domestic debt stock is low—gross domestic debt is estimated at approximately 3.5 percent of GDP at end-2007, and net debt at 0.6 percent of GDP. As a result, the baseline scenarios for external and total public debt are similar.

III. Conclusion

8. **Burkina Faso's risk of debt distress is high** on account of the NPV of debt-to-exports ratio; all other debt indicators remain comfortably below their policy-dependent thresholds under the baseline scenario. This risk underlines the need for a prudent fiscal policy, external borrowing limited to concessional loans, and continued build-up in public debt management capacity.

Appendix II—Table 1. Burkina Faso: External Debt Sustainability Framework, Baseline Scenario, 2008–28 1/

(percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2005	2006	2007			2008	2009	2010	2011	2012	2013	2008–13 Average	2018	2028	2014–28 Average
External debt (nominal) 1/	**38.9**	**21.1**	**19.8**			**21.4**	**24.4**	**26.8**	**28.1**	**29.0**	**29.8**		**32.5**	**32.7**	
o/w public and publicly guaranteed (PPG)	38.9	21.1	19.8			21.4	24.4	26.8	28.1	29.0	29.8		32.5	32.7	
Change in external debt	-2.9	-17.8	-1.2			1.5	3.1	2.4	1.3	0.9	0.8		0.1	-0.1	
Identified net debt–creating flows	8.0	6.7	0.2			8.1	8.4	8.5	7.7	6.7	6.2		3.7	3.5	
Non–interest current account deficit	**11.4**	**9.3**	**8.1**	**10.0**	**1.2**	**11.7**	**10.8**	**10.3**	**9.8**	**9.2**	**8.7**		**6.7**	**4.9**	6.0
Deficit in balance of goods and services	15.6	14.0	14.3			17.2	16.1	14.5	13.8	13.2	12.6		10.5	9.0	
Exports	9.9	11.5	10.5			10.2	11.3	11.7	11.8	11.8	11.9		11.5	11.7	
Imports	25.5	25.5	24.8			27.4	27.3	26.2	25.6	25.0	24.5		22.0	20.7	
Net current transfers (negative = inflow)	-4.3	-4.4	-6.0	-4.8	0.9	-5.5	-5.1	-4.0	-3.9	-3.9	-3.8		-3.7	-4.1	-3.8
o/w official	-3.4	-3.0	-4.3			-4.1	-3.7	-2.7	-2.5	-2.4	-2.3		-1.8	-1.6	
Other current account flows (negative = net inflow)	0.1	-0.2	-0.3			0.0	-0.1	-0.2	-0.1	-0.1	-0.1		-0.1	-0.1	
Net FDI (negative = inflow)	**-0.6**	**-0.6**	**-5.1**	**-0.9**	**1.5**	**-3.1**	**-1.7**	**-0.8**	**-1.0**	**-1.3**	**-1.3**		**-1.6**	**0.0**	-0.8
Endogenous debt dynamics 2/	**-2.8**	**-2.0**	**-2.8**			**-0.6**	**-0.7**	**-0.9**	**-1.2**	**-1.2**	**-1.3**		**-1.4**	**-1.3**	
Contribution from nominal interest rate	0.3	0.2	0.2			0.2	0.4	0.4	0.3	0.3	0.3		0.4	0.4	
Contribution from real GDP growth	-2.8	-2.0	-0.6			-0.7	-1.0	-1.3	-1.5	-1.5	-1.6		-1.8	-1.8	
Contribution from price and exchange rate changes	-0.3	-0.3	-2.4			…	…	…	…	…	…		…	…	
Residual (3–4) 3/	**-10.9**	**-24.6**	**-1.4**			**-6.6**	**-5.4**	**-6.2**	**-6.3**	**-5.8**	**-5.4**		**-3.6**	**-3.6**	
o/w exceptional financing	0.0	-23.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	19.7	11.2	12.0			12.6	14.8	16.2	17.4	18.2	19.1		21.0	21.7	
percent of exports	199.1	97.3	113.9			123.2	131.3	138.6	147.7	154.0	160.5		182.8	186.0	
NPV of PPG external debt	**19.7**	**11.2**	**12.0**			**12.6**	**14.8**	**16.2**	**17.4**	**18.2**	**19.1**		**21.0**	**21.7**	
percent of exports	**199.1**	**97.3**	**113.9**			**123.2**	**131.3**	**138.6**	**147.7**	**154.0**	**160.5**		**182.8**	**186.0**	
percent of government revenues	**154.2**	**86.1**	**88.0**			**93.2**	**108.5**	**114.4**	**117.0**	**118.4**	**119.2**		**115.4**	**118.5**	
Debt service-to-exports ratio (percent)	**8.3**	**6.2**	**6.5**			**5.9**	**6.7**	**6.6**	**5.9**	**6.5**	**6.9**		**9.2**	**11.9**	
PPG debt service-to-exports ratio (percent)	**8.3**	**6.2**	**6.5**			**5.9**	**6.7**	**6.6**	**5.9**	**6.5**	**6.9**		**9.2**	**11.9**	
PPG debt service-to-revenue ratio (percent)	**6.4**	**5.5**	**5.0**			**4.4**	**5.6**	**5.5**	**4.7**	**5.0**	**5.1**		**5.8**	**7.6**	
Total gross financing need (US$ billions)	631.1	548.6	248.4			743.3	798.6	900.5	910.2	900.6	931.8		1036.3	2337.9	
Non–interest current account deficit that stabilizes debt ratio	14.3	27.2	9.3			10.2	7.7	8.0	8.5	8.3	7.9		6.5	4.9	
Key macroeconomic assumptions															
Real GDP growth (percent)	7.1	5.5	3.6	5.6	1.9	4.5	4.9	6.0	6.0	6.0	6.0	5.6	5.9	5.9	5.9
GDP deflator in US$ terms (percent change)	0.8	0.7	12.8	5.3	10.3	13.8	-4.2	2.5	2.5	2.8	2.3	3.3	2.3	2.2	2.3
Effective interest rate (percent) 5/	0.7	0.7	1.2	0.6	0.4	1.1	1.7	1.8	1.1	1.2	1.2	1.3	1.3	1.5	1.3
Growth of exports of G&S (US$ terms, percent)	-2.1	23.7	6.9	12.3	22.2	15.5	10.8	12.9	9.2	9.5	8.9	11.1	7.1	9.5	8.2
Growth of imports of G&S (US$ terms, percent)	11.7	6.1	14.0	11.1	12.8	31.1	0.3	4.3	6.0	6.4	6.3	9.1	6.6	8.3	7.1
Grant element of new public sector borrowing (percent)	…	…	…	…	…	32.2	33.9	32.9	32.6	32.2	31.8	32.6	30.0	27.7	29.2
Aid flows (US$ billions) 7/	554.8	564.6	678.6			858.2	826.8	917.7	932.0	961.6	989.3		1077.3	2452.9	
o/w Grants	249.8	280.3	439.4			495.7	487.0	511.0	518.1	532.9	537.9		538.7	1226.5	
o/w Concessional loans	259.4	254.9	239.2			362.5	339.8	406.8	413.9	428.7	451.4		538.7	1226.5	
Grant–equivalent financing (percent of GDP) 8/	…	…	…			7.7	7.5	7.4	6.9	6.5	6.1		4.1	4.2	4.3
Grant–equivalent financing (percent of external financing) 8/	…	…	…			71.0	72.4	70.3	70.1	69.8	68.9		65.0	63.8	64.6
Memorandum items:															
Nominal GDP (US$ millions)	5436.9	5776.6	6753.0			8028.5	8071.2	8770.0	9526.9	10376.7	11254.4		16899.1	37351.5	
(NPVt–NPVt–1)/GDPt–1 (percent)						3.0	2.3	2.8	2.6	2.5	2.5	2.6	1.9	1.8	1.9

Source: Staff simulations.

1/ Includes both public and private sector external debt.

2/ Derived as [r – g – r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections, capital transfers are included–in particular project grants, which are projected to average about 2.5 percent of GDP over the long term, and about 4 percent over 2008-12 due to MCC grants—and private, non-debt-creating capital inflows. Projections also includes contribution from price and exchange rate changes.

4/ Assumes that NPV of private sector debt is equivalent to its face value.

5/ Current–year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant–equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Appendix II—Table 2. Burkina Faso: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2008–28
(Percent)

	Projections							
	2008	2009	2010	2011	2012	2013	**2018**	2028
NPV of debt-to-GDP ratio								
Baseline	13	15	16	17	18	19	**21**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 1/	13	13	14	15	16	18	**27**	37
A2. New public sector loans on less favorable terms in 2008–27 2/	13	16	19	21	23	24	**30**	34
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	13	15	17	18	19	20	**22**	22
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	13	16	20	21	22	23	**23**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	13	15	18	19	20	21	**23**	24
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	13	17	19	20	21	22	**23**	22
B5. Combination of B1–B4 using one–half standard deviation shocks	13	15	18	19	20	21	**22**	22
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	13	22	24	25	27	28	**31**	32
NPV of debt-to-exports ratio								
Baseline	123	131	139	148	154	160	**183**	186
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007–26 1/	123	118	121	128	138	149	**239**	318
A2. New public sector loans on less favorable terms in 2007–26 2/	123	142	159	177	192	206	**258**	293
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	123	131	139	148	154	160	**183**	186
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	123	177	268	279	287	295	**311**	292
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	123	131	139	148	154	160	**183**	186
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	123	152	166	174	179	184	**197**	189
B5. Combination of B1–B4 using one–half standard deviation shocks	123	154	192	202	209	216	**236**	230
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	123	131	139	148	154	160	**183**	186
NPV of debt-to-revenue ratio								
Baseline	93	108	114	117	118	119	**115**	119
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007–26 1/	93	98	99	101	106	111	**151**	202
A2. New public sector loans on less favorable terms in 2007–26 2/	93	117	131	141	148	153	**163**	187
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	93	110	118	121	122	123	**119**	123
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	93	119	143	143	143	142	**127**	121
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	93	109	125	127	129	130	**126**	129
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	93	125	137	138	137	137	**124**	120
B5. Combination of B1–B4 using one–half standard deviation shocks	93	111	129	130	131	131	**121**	119
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	93	158	167	170	172	174	**168**	173

continued

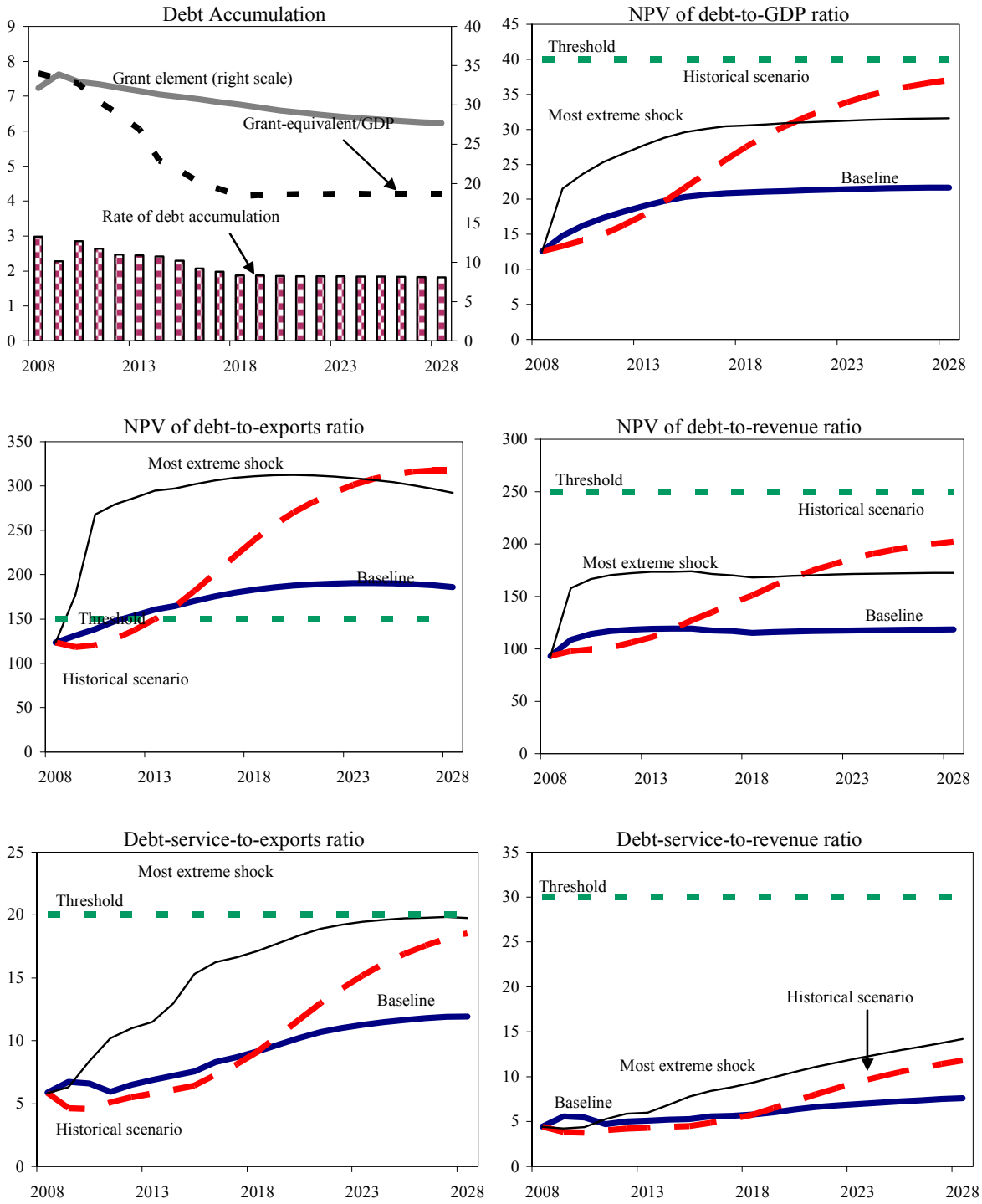
Appendix II. Table 2. Burkina Faso: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2008–28 (concluded)
(Percent)

	Projections							
	2008	2009	2010	2011	2012	2013	2018	2028
Debt service-to-exports ratio								
Baseline	6	7	7	6	7	7	**9**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 1/	6	5	5	5	6	6	**9**	19
A2. New public sector loans on less favorable terms in 2008–27 2/	6	5	5	7	8	8	**15**	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	6	5	5	6	7	7	**9**	12
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	6	6	8	10	11	12	**17**	20
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	6	5	5	6	7	7	**9**	12
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	6	5	6	6	7	7	**11**	13
B5. Combination of B1–B4 using one–half standard deviation shocks	6	6	7	8	8	9	**12**	15
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	6	5	5	6	7	7	**9**	12
Debt service-to-revenue ratio								
Baseline	4	6	5	5	5	5	**6**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 1/	4	4	4	4	4	4	**6**	12
A2. New public sector loans on less favorable terms in 2008–27 2/	4	4	4	5	6	6	**9**	14
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	4	4	4	5	5	5	**6**	8
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	4	4	4	5	5	6	**7**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	4	4	5	5	5	6	**6**	8
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	4	4	5	5	5	5	**7**	8
B5. Combination of B1–B4 using one–half standard deviation shocks	4	4	4	5	5	5	**6**	8
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	4	6	6	7	7	7	**8**	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	33	33	33	33	33	33	**33**	33

Source: Staff projections and simulations.

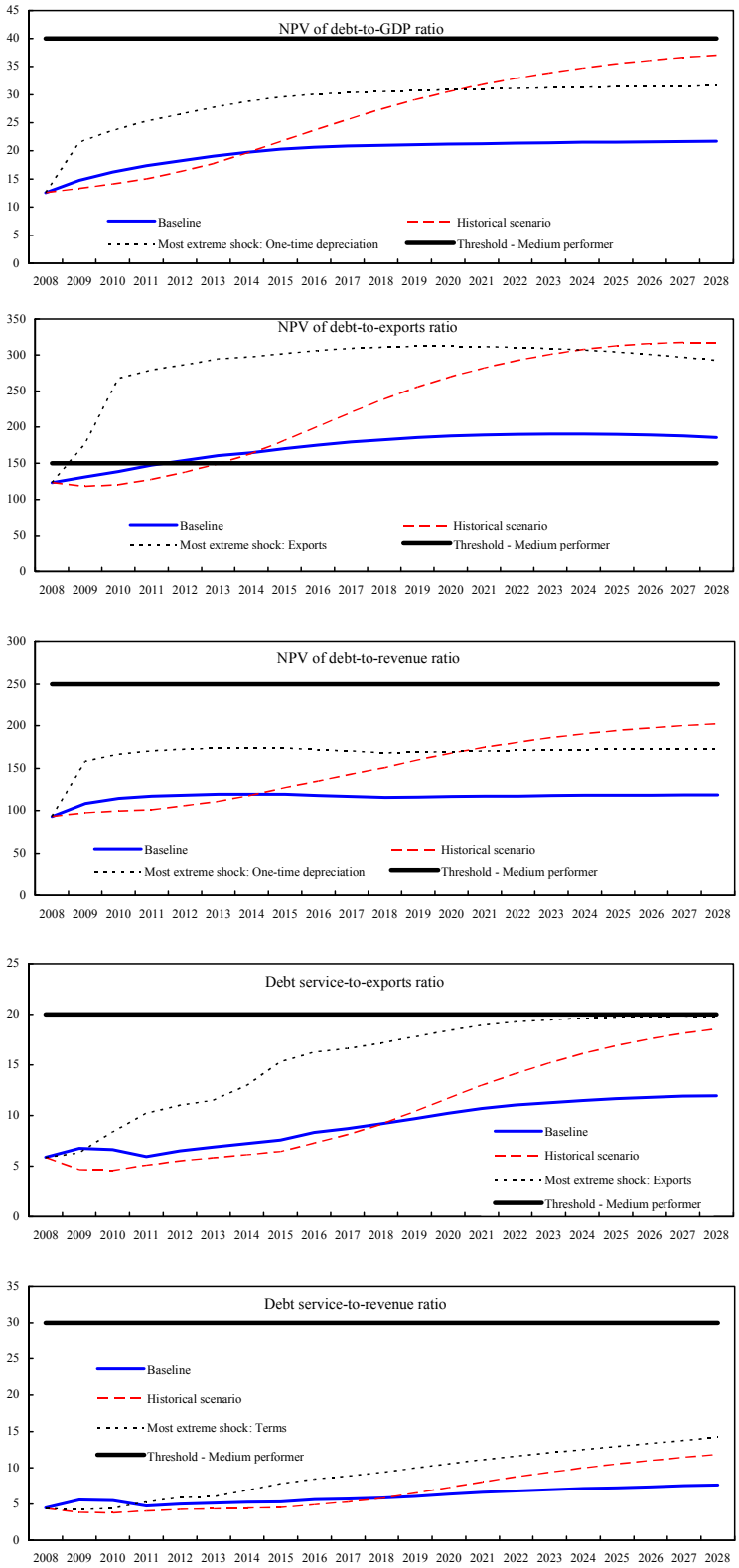
1/ Variables include real GDP growth, growth of GDP deflator (US$ terms), non–interest current account in percent of GDP, and non–debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assumingan offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Appendix II—Figure 1. Burkina Faso: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios (in percent), 2008-28 1/



Source: Staff projections and simulations.
1/ Thresholds for medium performer.

Appendix II. Figure 2. Burkina Faso: Indicators of Public External Debt in Alternative Scenarios, 2007–28 (Percent)



Source: IMF staff projections and simulations.

Appendix II–Table 3.Burkina Faso: Public Sector Debt Sustainability Framework, Baseline Scenario, 200528

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2005	2006	2007	Historical Average 5/	Standard Deviation 5/	2008	2009	2010	2011	2012	2013	2008-13 Average	2018	2028	2014-28 Average
Public sector debt 1/	44.4	22.8	22.1			24.9	26.0	28.0	28.6	29.9	31.1		33.1	32.2	
o/w foreign-currency denominated	40.9	20.0	18.6			23.1	24.4	26.7	28.0	28.9	29.7		32.4	32.7	
Change in public sector debt	0.3	-21.6	-0.7			2.8	1.1	2.0	0.6	1.3	1.2		0.0	-0.1	
Identified debt-creating flows	6.6	-23.0	2.4			4.4	2.8	2.3	1.7	1.3	1.2		0.0	-0.1	
Primary deficit	4.6	5.0	5.3	3.6	1.2	4.9	4.3	3.8	3.5	3.3	3.1	3.8	2.1	1.9	2.1
Revenue and grants	17.4	18.9	20.1			19.7	19.7	20.0	20.3	20.5	20.8		21.4	21.6	
of which : grants	4.6	5.9	6.5			6.2	6.0	5.8	5.4	5.1	4.8		3.2	3.3	
Primary (noninterest) expenditure	21.9	23.9	25.4			24.6	24.0	23.8	23.8	23.8	23.9		23.5	23.4	
Automatic debt dynamics	2.1	-5.9	-2.9			-0.5	-1.5	-1.5	-1.8	-1.9	-1.9		-2.1	-2.0	
Contribution from interest rate/growth differential	-3.6	-3.1	-1.0			-1.1	-1.1	-1.3	-1.7	-1.7	-1.8		-2.0	-1.9	
of which : contribution from average real interest rate	-0.7	-0.8	-0.2			-0.2	0.1	0.2	-0.1	-0.1	-0.1		-0.1	-0.1	
of which : contribution from real GDP growth	-2.9	-2.3	-0.8			-0.9	-1.2	-1.5	-1.6	-1.6	-1.7		-1.9	-1.8	
Contribution from real exchange rate depreciation	5.7	-2.8	-1.9			0.6	-0.5	-0.2	-0.2	-0.2	-0.1		...	...	
Other identified debt-creating flows	0.0	-22.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	-22.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-6.3	1.4	-3.1			-1.6	-1.7	-0.3	-1.1	0.0	0.0		0.0	0.0	
NPV of public sector debt	23.2	14.0	15.4			14.4	16.4	17.5	17.9	19.2	20.4		21.7	21.3	
o/w foreign-currency denominated	19.7	11.2	12.0			12.6	14.8	16.2	17.4	18.2	19.1		21.0	21.7	
o/w external	19.7	11.2	12.0			12.6	14.8	16.2	17.4	18.2	19.1		21.0	21.7	
NPV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need 2/	6.0	6.2	6.6			5.7	5.2	4.7	4.3	4.2	4.0		3.2	3.3	
NPV of public sector debt-to-revenue and grants ratio (in percent)	133.6	73.9	76.7			73.1	83.5	87.4	88.3	93.5	98.1		101.5	98.5	
NPV of public sector debt-to-revenue ratio (in percent)	181.7	107.3	113.3			106.4	120.4	123.3	120.7	124.7	127.5		119.3	116.1	
o/w external 3/	154.2	86.1	88.0			93.2	108.5	114.4	117.0	118.4	119.2		115.4	118.5	
Debt service-to-revenue and grants ratio (in percent) 4/	8.2	6.2	6.7			4.0	4.7	4.6	4.1	4.4	4.5		5.3	6.6	
Debt service-to-revenue ratio (in percent) 4/	11.2	8.9	9.9			5.8	6.7	6.5	5.6	5.8	5.8		6.2	7.8	
Primary deficit that stabilizes the debt-to-GDP ratio	4.3	26.6	6.0			2.1	3.2	1.8	2.9	2.0	1.9		2.1	2.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.1	5.5	3.6	5.6	1.9	4.5	4.9	6.0	6.0	6.0	6.0	5.6	5.9	5.9	5.9
Average nominal interest rate on forex debt (in percent)	0.7	0.6	1.1	0.6	0.4	1.1	1.7	1.8	1.1	1.2	1.2	1.3	1.3	1.5	1.4
Average real interest rate on domestic currency debt (in percent)	3.9	5.1	4.1	-0.3	5.2	0.5	6.1	7.7	9.6	22.3	10.5	9.4	7.0	-12.0	-2.5
Real exchange rate depreciation (in percent, + indicates depreciation)	16.1	-7.3	-10.0	-2.5	12.5	3.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.6	-0.1	3.4	2.8	2.9	5.1	3.4	2.1	2.1	2.3	2.3	2.9	2.3	2.2	2.3
Growth of real primary spending (deflated by GDP deflator, in percent)	10.8	15.0	10.2	7.4	6.6	1.0	2.3	5.4	5.8	6.0	6.4	4.5	5.9	5.9	5.8
Grant element of new external borrowing (in percent)	0.0	0.0	0.0	0.0	0.0	32.2	33.9	32.9	32.6	32.2	31.8	32.6	30.0	27.7	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ Central government.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

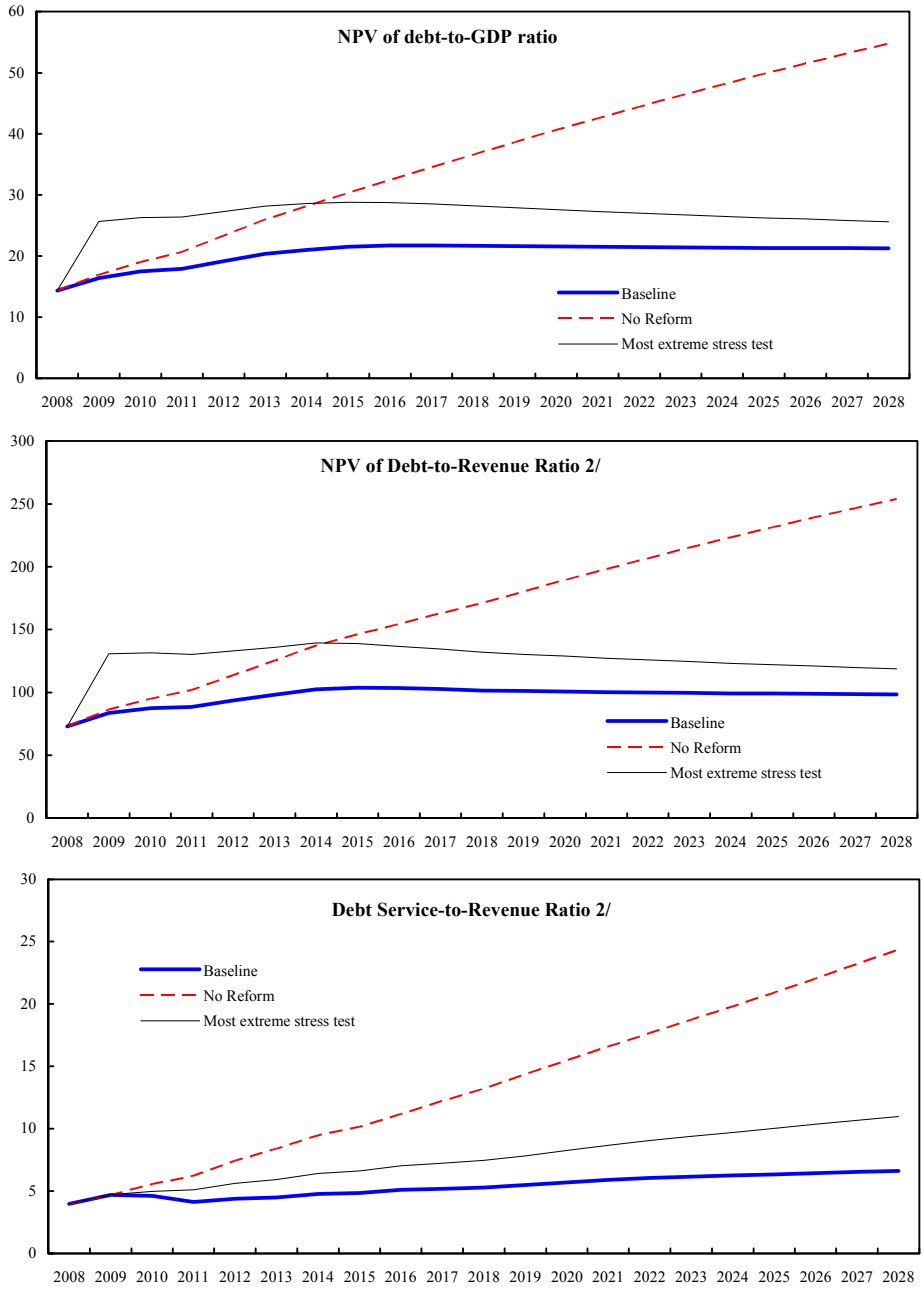
Appendix II–Table 4.Burkina Faso: Sensitivity Analysis for Key Indicators of Public Debt 2008-28

	Projections							
	2008	2009	2010	2011	2012	2013	2018	2028
NPV of Debt-to-GDP Ratio								
Baseline	14	16	17	18	19	20	22	21
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	14	16	17	17	19	21	27	38
A2. Primary balance is unchanged from 2008	14	17	19	21	23	26	37	55
A3. Permanently lower GDP growth 1/	14	17	18	18	20	22	26	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	14	17	19	20	21	23	27	30
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	14	17	19	19	20	22	23	22
B3. Combination of B1-B2 using one half standard deviation shocks	14	16	18	18	19	21	22	21
B4. One-time 30 percent real depreciation in 2009	14	22	22	21	22	23	22	21
B5. 10 percent of GDP increase in other debt-creating flows in 2009	14	26	26	26	27	28	28	26
NPV of Debt-to-Revenue Ratio 2/								
Baseline	73	83	87	88	94	98	102	98
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	73	80	83	85	92	99	125	176
A2. Primary balance is unchanged from 2008	73	86	95	102	114	125	171	254
A3. Permanently lower GDP growth 1/	73	84	89	91	97	104	120	159
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	73	85	92	96	103	110	124	137
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	73	86	94	95	100	104	106	102
B3. Combination of B1-B2 using one half standard deviation shocks	73	83	89	90	94	98	100	95
B4. One-time 30 percent real depreciation in 2009	73	110	108	106	108	110	104	96
B5. 10 percent of GDP increase in other debt-creating flows in 2009	73	131	131	130	133	136	132	119
Debt Service-to-Revenue Ratio 2/								
Baseline	4	5	5	4	4	4	5	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	4	5	4	3	4	5	8	15
A2. Primary balance is unchanged from 2008	4	5	6	6	7	8	13	24
A3. Permanently lower GDP growth 1/	4	5	5	4	5	5	7	13
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	4	5	5	5	6	6	7	11
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	4	5	5	6	5	5	6	7
B3. Combination of B1-B2 using one half standard deviation shocks	4	5	4	5	4	4	5	6
B4. One-time 30 percent real depreciation in 2009	4	5	6	5	5	6	7	9
B5. 10 percent of GDP increase in other debt-creating flows in 2009	4	5	21	10	8	7	7	10

Sources: Country authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Appendix II. Figure 3.Burkina Faso: Indicators of Public Debt Under Alternative Scenarios, 2008-28 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2018.
2/ Revenue including grants.